KAOSAMPLE User Guide

A feature layer for the stock MPC Sample app (firmware 1.3.0.12)

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KAOSAMPLE v0.32 User Guide

Tester beta. Private copy, please do not share or repost.

Where this guide says **(check on device)**, the written notes for this build did not settle the exact button, label or behavior. Look at the screen, and tell Omar what you see.

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1. What it is, safety notes and the beta warning

1.1 What KAOSAMPLE is

KAOSAMPLE adds tools on top of the stock Akai MPC Sample app. Everything the MPC does today still works. KAOSAMPLE adds new pages, new tools and its own master processing (EQ, Loudness, Color).

It runs from the SD card and lives in the MPC's memory only. It does not change anything inside the MPC itself. Take the card out, or delete the `autorun` folder from the card, and your MPC is 100% stock again.

Made by Omar Huertas LLC / kaosbeatz. It uses its own code and its own artwork.

1.2 What you need

Need	Detail
MPC Sample	Firmware **1.3.0.12** exactly. On any other firmware KAOSAMPLE turns itself off and the MPC runs stock.

Need	Detail
SD card	In the MPC, with free space. Some tools write big files (stems, templates, loops).
A computer	To copy the `autorun` folder onto the card and to read the logs if something goes wrong.
A MIDI keyboard (optional)	Only for the “Keys” mode in Perform.

1.3 Beta warning (please read)

- **This is a beta. It has not been fully tested on a real MPC. It may crash.**
- The honest test status of each tool is in the table below. Everything from v0.21 up was tested in an emulator (a computer simulation of the MPC) and **has not run on a real unit yet**. Your testing on the real machine is the point of this build.
- **Back up your SD card first.** Copy the whole card to your computer before you install.
- **Crash fallback:** if KAOSAMPLE crashes twice within 2 minutes, it gives up and starts the stock app on its own. You will see the normal MPC with none of the KAOSAMPLE tools. Wait a few minutes and restart the MPC to try again.
- **Lost project?** KAOSAMPLE saves a copy of your project every 5 minutes into `KAOSAMPLE/snapshots` on the card. See chapter 4.5 (Snapshots and recovery).

Test status by tool

Tool	Version	Status
Master EQ and spectrum, overwrite-record fix	0.15	Tested on hardware
Undo history, snapshots, system monitor, explode, load kit from card	0.16	Tested on hardware
Step edit (copy, paste, bar menu, zoom)	0.17	Tested on hardware
Track tools (solo, rename, move, delete, mute scenes)	0.18	Tested on hardware
Sample tools (chop, extract, reverse, tail, fade)	0.19	Tested on hardware
Pad sound editing	0.20	Tested on hardware
Pad inserts (FX) and Loudness	0.21	Emulator only
Stems and Song to Seq	0.22	Emulator only
Auto-kit	0.24	Emulator only
Groove	0.25	Emulator only
Retro Rec, tap tempo, Save version	0.26	Emulator only
Clean up (purge, mono)	0.27.1	Emulator only

Tool	Version	Status
Perform (scale, chord, arp, Keys)	0.28	Emulator only
Looper (Loop Rec)	0.29	Emulator only
Files and templates	0.30 / 0.30.1	Emulator only
Color, Duck, Reference A/B	0.31	Emulator only
Duck sources, Color smoothing and undo memory, Looper line-in / free-run / tempo stretch / export, Perform Arp + Keys, Gate, Pad MIDI In at boot, Files (Jump in FX list, default template, Starter STRIP), Sample Record undo, project sidecar	0.32	Emulator only

1.4 Safety notes

- **Back up the card first** (see above). Back up your important projects to your computer as well.
- **Do not rename or edit** the files in the `autorun` folder. The MPC checks them.
- **Delete track cannot be undone.** Use Snapshots to get a deleted track back.
- **Record a sample: undo works on the Sample Record screen** (SHIFT + MINUS / SHIFT + PLUS). Undo waits about 1.2 s after you stop recording, so the take can settle first.
- **Stock insert effects on tracks and master do nothing** on this unit (a limit of the MPC Sample, not a bug). Use KAOSAMPLE's own master EQ, Loudness and Color instead. Insert effects on **pads** do work.
- **Two quick restarts = stock app.** KAOSAMPLE restarts the MPC app itself for a few jobs (restoring a snapshot, New from template). If the app exits twice less than 2 minutes apart, KAOSAMPLE switches itself off and the stock app runs. KAOSAMPLE refuses those jobs until the app has been up for about 2 minutes.
- **Some data lives on the card, not only in the project.** Grooves and the Color settings are kept on the card. Mute scenes, recorded mutes, slice sets and chord sets are saved with the project (a sidecar file, see 4.7) and also kept on the card, and they come back when the project loads. If you move a project to another card, the card copies do not come along.

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2. Install, update and uninstall

2.1 Install

1. Power off the MPC. Take the SD card out and put it in your computer.

2. **Back up the card** (copy everything to a folder on your computer).

3. Copy the **`autorun`** folder from the zip to the **top level** of the card. The card should then have:

   ```
   /autorun/autorun.json
   /autorun/autorun.sh
   /autorun/autorun.check
   /autorun/kaos-run.sh
   /autorun/kaos-mem.sh
   /autorun/kaoslayer.so
   /autorun/kaosplash
   ```

   Do not rename or edit any file.

4. Eject the card safely and put it in the MPC.

5. Turn the MPC on. It starts KAOSAMPLE by itself. You will see the KAOSAMPLE splash, then the normal MPC screen.

6. A **`KAOSAMPLE`** folder appears on the card. Your logs and your KAOSAMPLE files live there.

2.2 Check that it is running

- You saw the KAOSAMPLE splash at start-up.
- **SHIFT + SCREEN 3** opens the KAOSAMPLE menu (a grid of tiles).
- A `KAOSAMPLE` folder exists on the card.

If you see only the normal MPC and no KAOSAMPLE menu, see chapter 6 (Troubleshooting).

2.3 Update to a new build

The tester notes only describe a fresh install. The simple way to update is the same as installing:

1. Power off the MPC and put the card in your computer.
2. Delete the old `autorun` folder from the card.
3. Copy the new `autorun` folder to the top of the card.
4. Leave the `KAOSAMPLE` folder alone. It holds your snapshots, slices, grooves, templates and settings.
5. Put the card back and start the MPC.

(check on device) that your settings and snapshots are still there after the first start of a new build.

2.4 Uninstall

- Delete the **`autorun`** folder from the card, **or** just start the MPC without the card.
- Your MPC is stock again.
- You can keep or delete the `KAOSAMPLE` folder. It is only your data and logs.

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3. Getting around

3.1 The controls this guide uses

Name in this guide	What it is
K1, K2, K3	The three knobs. On a KAOSAMPLE page they edit the three values of the selected row.
S1, S2, S3	The **SCREEN 1, SCREEN 2, SCREEN 3** buttons. On a KAOSAMPLE page they run the actions named in the top bar and in the hint line.
DATA	The data wheel. **Turn** it to move up and down. **Push** it to press it in.
ERASE	The ERASE button.
SHIFT	The SHIFT button, held while you press another control.
Pads and PAD BANK	The 16 pads. PAD BANK A to H gives you 128 pads.
PLUS / MINUS	The PLUS and MINUS buttons.

3.2 Opening the KAOSAMPLE menu

Press **SHIFT + SCREEN 3**. A grid of tiles opens. Pick a tile and open it. (check on device) which controls move the highlight and open a tile; the notes do not spell this out.

The menus are linked like this:

1. The first menu, titled **KAOSAMPLE**.
2. **KAOS TOOLS** (the second menu). The tester notes say to open “KAOS TOOLS” from the menu, but the first menu has no tile with that name, so (check on device) which tile or button opens KAOS TOOLS.
3. **KAOS TOOLS 2** (the third menu). Open it with the **More** tile on KAOS TOOLS. The **Back** tile on KAOS TOOLS 2 returns.

In short: **SHIFT + SCREEN 3 > KAOS TOOLS > More > KAOS TOOLS 2**.

First menu: KAOSAMPLE

Mixer, Pad Mixer, Tracks, Track Mute, Sequences, Song, Track View, Pad View, Record, Theme, Meter, Header, About, Close, and **More** (opens KAOS TOOLS). The old **iPad** tile was removed in v0.31.1: it only showed an address and did nothing.

Four tiles are switches, not pages. Push one and it flips at once; the menu stays open. Each switch shows a **1.5 second message** in the middle of the screen, and a switch that is **On** is drawn with a bright border and a green marker in the tile corner. The big info box at the bottom right of the menu shows the state of the tile you are on.

Switch	What it does	Message
Record	Chooses what a plain **SEQ REC** press does. **Overwrite** (default): REC erases the old notes the loop passes over, and **SHIFT + REC** adds without erasing. **Overdub**: REC adds without erasing, and **SHIFT + REC** erases. The setting is read at the next REC press.	"Record: Overdub" or "Record: Overwrite"
Theme	Each push steps to the next colour theme (Stock, MPC60, MPC3000, MPC2000XL, MPC1000, MPC Live). It tints the whole screen, the stock screens too.	"Theme" and the theme name
Meter	A small output level bar at the bottom right of every screen (green, red near the top; it holds the peak for about 0.3 s). It follows the MPC's main output.	"Meter: On" or "Meter: Off"
Header	A one-line strip "S01 T01 A01" (sequence, track, pad) under the tabs of the stock screens. It is only drawn where that strip is empty, and only when the project has at least one track. Default is On.	"Header: On" or "Header: Off" (when the project has no tracks: "shows when a project has tracks")

These four switches are not remembered after a restart (Header starts On, the rest start Off or Overwrite).

This guide covers **Track View** and **Pad View** (4.6), **Tracks** and **Track Mute** (4.7), **Mixer** (explode, 4.8) and **About** (monitor, 4.20). The others (Pad Mixer, Sequences, Song) are not described in the notes for this build, so they are not covered here.

KAOS TOOLS (second menu)

Tile	What it opens	Chapter
Master	(check on device)	
Pad Sound	Envelopes, filter, choke, layers for one pad	4.10
Seq Ops	Sequence tools, includes Explode	4.8
MIDI	MIDI page (not covered)	
Looper	The older looper page that drives the stock looper	4.18
Stock screens	Shortcuts to stock MPC screens (not covered)	
Programs	Programs page (not covered)	
EQ Spectrum	Master EQ and spectrum	4.1
Undo History	Named list of what you did	4.4
Snapshots	Restore a saved project copy	4.5
FX	Pad insert effects	4.10
Loudness	Master loudness	4.2
Tracks Pro	Solo, rename, move, delete	4.7
Mute Scenes	Save and recall mute patterns	4.7
Sample Tools	Chop, extract, reverse, tail, fade	4.9
More	Opens KAOS TOOLS 2	

KAOS TOOLS 2 (third menu)

Tile	What it opens	Chapter
Stems	Export every track and the master as WAV files	4.11
Song to Seq	Turn a song into one sequence	4.12
Groove	Swing, humanize and groove templates	4.14
Auto-kit	Build a 16-pad kit automatically	4.13
Retro Rec	Grab what you just played; tap tempo; Save version	4.15
Perform	Scales, chords, arp, Keys	4.17
Clean Up	Purge unused samples, make mono	4.16
Loop Rec	The new multi-pass looper	4.18
Files	Templates, new folders, random names, browser jump	4.19
Color	Tape and vinyl color, duck, reference A/B	4.3
Back	Back to KAOS TOOLS	

The two menus have some empty tiles. They do nothing.

3.3 Page conventions: K1-K3, S1-S3, DATA and ERASE

Most KAOSAMPLE pages (Color, Perform, Groove, Retro Rec, Clean Up, Files, Looper, Auto-kit, Pad Sound) follow the same habits.

Control	What it does on a KAOSAMPLE page
DATA turn	Picks the **row** in the list on the page.
K1, K2, K3	Edit the **three values of the selected row**. The names of K1, K2, K3 for that row show at the bottom of the screen.
S1, S2, S3	Run actions. The names show in the top bar (left, middle, right) and in the hint line. They change with the row.
DATA push	Runs the row's main action (load, run, open a menu), when the page has one.
ERASE	Clears or cancels something small. See each page.
ERASE twice ("ERASE x2")	Confirms a bigger clear or delete. Press it again within a few seconds (about 4 seconds on Tracks and Looper). The first press shows a message such as "ERASE again".
SHIFT + DATA turn	On many pages, picks the **previous or next track**. On Pad Sound it picks the previous or next pad.
SHIFT + ERASE	A reset on several pages (EQ, Loudness, Color, Name editor). On the grids and mixer it is **Explode**.
Pads	Keep playing on every page. A few pages also use pads as a keyboard or selector.

Good to know:

- **Knobs jump.** On Color and Loudness the knobs are absolute: the value jumps to where the knob is.
- **Pad Sound has "soft takeover"**: after you change row or pad, the knob must come close to the shown value before it takes over. You see the message "Knob: move to the shown value to pick it up".
- **A message line** appears for a few seconds at the bottom of the page. Read it, because refusals are explained there.
- **Doing two-press things.** "S2 twice" or "S3 twice" means press the same button twice within the time shown on screen (6 seconds for Purge, 5 seconds for Snapshots).

3.4 The SHIFT table

Hold **SHIFT** on a KAOSAMPLE page and an overlay lists the SHIFT combos for that page. It disappears when you let go. It never delays or blocks a combo. It is the quickest way to check a control.

Everywhere

Combo	Action
SHIFT + SCREEN 3	Open the KAOSAMPLE menu
SHIFT + MINUS	Undo (stock)
SHIFT + PLUS	Redo (stock)
SHIFT + SAMPLE RECORD	Retro Rec: drop what you just played into the current track (chapter 4.15)
SHIFT + TAP TEMPO	Tap tempo (plain TAP TEMPO is still the stock button)
PAD FX, pressed twice	Mixer pages (tabs: SCREEN 1 Pad Mixer, SCREEN 2 Channel Mixer, SCREEN 3 track and pad grid)
PAD BANK, hold	Tracks grid (a pad selects that track)
SHIFT + PAD BANK, hold	Track Mute grid (a pad toggles mute for that track)
SHIFT + PAD BANK, tap	Create a new track
PAD BANK or PAD FX, single tap	The stock function, as normal

The “PAD FX x2”, “PAD BANK” and “SHIFT + PAD BANK” combos copy habits Omar reported from other MPC add-ons so your muscle memory carries over.

By page

Page	SHIFT combos
Mixer, Track View, Pad View, Seq Ops, Tracks Pro	SHIFT + ERASE = Explode the current track
Pad Mixer	SHIFT + pad = mute that pad
Track View, Pad View	SHIFT + DATA push = select the whole row. SHIFT + DATA turn = move the row. SHIFT + K1 / K2 = nudge / length in steps of 1/16. SHIFT + CHOP = Cut. SHIFT + LOOP = Duplicate
EQ Spectrum	SHIFT + ERASE = reset EQ flat (all gains 0, all kills off)
Loudness	SHIFT + ERASE = turn Loudness off
Color	SHIFT + ERASE = reset Color, Duck and Trim to defaults (undoable)

Page	SHIFT combos
Tracks Pro	SHIFT + DATA turn = move the track. SHIFT + pad = mute that track
Name editor	SHIFT + pad = lower case. SHIFT + ERASE = clear the name
Mute Scenes	SHIFT + pad (bottom two rows) = store scene. SHIFT + pad (top two rows) = solo. SHIFT + ERASE = clear last scene
Pad Sound	SHIFT + DATA turn = previous or next pad
Slice view	SHIFT + DATA turn = destination start pad. SHIFT + pad = extract the selected slice to that pad
FX list	SHIFT + ERASE = clear the name filter
Groove, Retro Rec, Clean Up, Looper	SHIFT + DATA turn = previous or next track
Files (in stock Sample Select, Project Select and FX Select)	SHIFT + DATA turn = jump 5, 10 or 20 rows
Sample Record screen	SHIFT + MINUS / SHIFT + PLUS = undo / redo a sample recording (once it has settled)
Programs	SHIFT + DATA push = Save kits (not covered here)

4. The features

Each feature has the same four parts: **What it does**, **Step by step**, **Controls** and **Tips and limits**. Test status for each is in chapter 1.3.

4.1 Master EQ, kill EQ and spectrum

What it does

A 3-band equalizer on the **master output**: low, mid and high, each from minus 12 to plus 12 dB. Each band also has a **kill** switch that silences just that band (handy for DJ-style drops). A 24-band spectrum analyzer shows what your mix is doing. Settings are saved on the card and come back after a restart.

Step by step

1. **SHIFT + SCREEN 3 > KAOS TOOLS > EQ Spectrum.**
2. Play your sequence.
3. Move the band controls to shape the master. Watch the spectrum.

4. Use a band's kill to drop it out, press again to bring it back.
5. **SHIFT + ERASE** resets everything flat.

Controls

Control	Action
K1, K2, K3	Band gains (low, mid, high). (check on device) which knob is which band
Kill buttons	One kill per band. (check on device) which buttons
SHIFT + ERASE	Reset EQ flat: all gains 0, all kills off

Tips and limits

- The EQ sits on the **master only**. There is **no per-track kill EQ**: the MPC Sample does not give KAOSAMPLE separate audio for each track.
- The EQ costs very little processing (well under 1% of the audio time budget in the emulator).
- The EQ is not in the stock undo list.

4.2 Loudness

What it does

A mastering chain on the master output, after the EQ. It has:

- a **3-band compressor** (low, mid, high),
- **saturation** (warmth and drive),
- a **lookahead brickwall limiter** with the ceiling at **-0.3 dBFS**,
- presets **Off, Glue, Punch, Loud, Max** and a one-knob **Loud** macro,
- a **level-matched A/B** so you compare the processed sound fairly (the unprocessed side is not made quieter than it is; the processed side is never boosted past the ceiling).

Settings are saved on the card.

Step by step

1. **SHIFT + SCREEN 3 > KAOS TOOLS > Loudness.**
2. Play your sequence.
3. Press **S3** to step through presets, or pick the **Main** row and turn **K1** (Preset) or **K2** (Loud).
4. Press **S1** to flip between processed and bypassed. Press **S2** to turn level match on, so both sound equally loud.
5. Watch the gain-reduction bars (Low, Mid, High and the limiter) and the output peak against the ceiling.
6. **SHIFT + ERASE** turns Loudness off.

Controls

DATA turn picks the row (check on device). K1, K2, K3 edit the row's three values.

Row	K1	K2	K3
Main	Preset	Loud (macro)	Push (extra gain into the limiter)
Low	Threshold	Ratio	Makeup
Mid	Threshold	Ratio	Makeup
High	Threshold	Ratio	Makeup
Time	Attack	Release	Detect (RMS or peak)
Sat / AB	Drive	Sat	LvlMatch

Button	Action
S1	A/B: processed or bypassed
S2	Level match on or off
S3	Next preset
SHIFT + ERASE	Loudness off

Tips and limits

- Presets get louder in this order: Glue (gentle), Punch, Loud, Max. Max is not much louder than Loud, because the ceiling is already reached; the extra push goes into limiting.
- The ceiling holds. In the emulator test no sample went over -0.3 dBFS in a 10-minute stress run.
- Bypass is a short crossfade, so switching does not click.
- Loudness is not in the stock undo list.
- Offline stem renders skip the EQ and Loudness. See chapter 4.11.

4.3 Color, Duck and Reference A/B

All three live on one page: **KAOS TOOLS 2 > Color**.

What it does

Color gives the master output tape or vinyl character with one **Mix** knob.

- **Tape:** saturation, head bump (low end warmth), high roll-off, wow and flutter (slow and fast pitch wobble).
- **Vinyl:** warmth, mono below about 120 Hz (a tighter low end), and crackle (random clicks).
- **Mix at 0 is exactly off and costs nothing.** Mix 0 means the sound is bit-for-bit unchanged.

Duck (sidechain) dips the output when a chosen track, a set of tracks or a pad plays, like a kick pumping a pad. MIDI-in notes and arp hits can trigger it too. Depth up to 24 dB, attack 0.1 to 50 ms, release 10 to 1000 ms, three curve shapes.

Reference A/B plays a WAV from your own music next to your mix, **matched by loudness** (a LUFS-style measure) so louder does not fool you into thinking it sounds better. A trim knob lets you nudge the match.

Where it sits in the signal path: Color goes before Loudness (so the limiter ceiling still holds). Duck and Reference go after the master volume.

Step by step: Color

1. Open **KAOS TOOLS 2 > Color**.
2. Pick the **Master** row. Turn **K2** up (upper half) to turn Tape on. Turn **K3** up to turn Vinyl on.
3. Raise **K1 (Mix)** until you hear it.
4. Pick the **Tape**, **Wow/Flut** or **Vinyl** rows and set the amounts.
5. **S2** undoes your last change. **S3** redoes it.

Step by step: Duck

1. Pick the **Duck src** row. Set **K1 (Source)** to **Track** (any note of that track ducks), **Pad** (one pad on that track) or **Set** (any note of any track in the set ducks). For Track and Pad, pick the track on **K2** and the pad on **K3**. For Set, K2 moves the cursor along the tracks and K3 puts the cursor's track in or out of the set.
2. Pick the **Duck env** row. Set **Depth, Attack, Release**. Pick the curve on the **Shape** row.
3. Push **DATA** on a duck row to fire a test duck so you can hear it without playing.
4. Watch the "duck" meter on the page.
5. Optional: on the **Duck in** row, **K1** adds MIDI-in notes (played on the current track, which must be in the set), **K2** adds arp and Perform hits, and **K3** sets all or none of the tracks in the set.

Step by step: Reference A/B

1. Copy WAV files into **`KAOSAMPLE/refs`** on the card (do this with the card in your computer).
2. Open **Color** and pick the **Ref** row.
3. Turn **K1** to choose a file. Push **DATA** to load it. The page says "loading" and then "ready".
4. Press **S1** to switch between **A: MPC** (your mix) and **B: REF** (the reference).
5. If the match sounds off, nudge **K2 (Trim)** between -6 and +6 dB.

Controls

Row	K1	K2	K3
Master	Mix	Tape on / off	Vinyl on / off
Tape	Sat	Bump	HF (high roll-off)
Wow/Flut	Wow	Flutter	-

Row	K1	K2	K3
Vinyl	Warmth	Crackle	Mono below 120 Hz on / off
Duck src	Source (Off, Track, Pad, Set)	Track (K2 moves the cursor in Set)	Pad, or in / out of the set (Set)
Duck in	MIDI in on / off	Arp / Perform on / off	All or none of the set
Duck env	Depth	Attack	Release
Shape	Linear, Exp, S-curve	-	-
Ref	File	Trim	-

For on / off knobs, the upper half of the knob is ON.

Button	Action
S1	A/B: your mix or the reference
S2	Undo
S3	Redo
DATA turn	Pick the row
DATA push	On the Ref row: load the file. On a duck row: test duck
SHIFT + ERASE	Reset Color, Duck and Trim (A/B returns to your mix). It is undoable

Tips and limits

- **Setting is saved** and comes back after a restart. The **reference file name is remembered but never loaded on its own**, and A/B **always starts on your mix**.
- **Undo/redo (32 steps) is saved too** (in `KAOSAMPLE/fxundo.txt`) and comes back after a restart.
- Duck sources: the notes of a set of tracks (muted tracks do not duck), live pad hits on a track in the set, MIDI-in notes (played on the current track) and arp / Perform hits, each if its switch is on.
- Duck from the sequence reacts within about 2.6 ms of the true note time. Live pad hits react within 2.9 ms.
- The loudness meter is **LUFS-style, not a certified meter.** It needs a few seconds of music to settle. Until then the reference plays at the trim only.
- If the reference is very loud and would clip, its level is capped and the page says **"peak cap"**. The match is then not exact.
- If you move the master volume, the matched reference follows slowly.
- The reference list shows up to 48 files. WAV only: 16, 24 or 32-bit PCM, or 32-bit float. Mono or stereo, any sample rate.
- Color settings glide to their new values (about 12 ms; wow and flutter about 60 ms), so fast knob sweeps no longer click. A very fast sweep is softened on purpose. The dry and wet signals stay aligned at any Mix.
- Stem renders include Color but **not** the duck or the reference (chapter 4.11).

4.4 Undo, redo and the history list

What it does

- A **popup** shows on every page when you undo or redo, for example “Undo 2/3”, with the name of the step.
- The **Undo History** page is a named list of your steps. Newest is on top. You can jump to any step.
- Nearly every KAOSAMPLE edit is a normal stock undo step. Several multi-part jobs (paste, quantize, a bar insert, a groove apply, an auto-kit) are **one** undo step.

Step by step

1. Press **SHIFT + MINUS** to undo and **SHIFT + PLUS** to redo, on any page. The popup shows where you are.
2. For the list, open **KAOS TOOLS > Undo History**.
3. Turn **DATA** to scroll. The green bar is the last step applied. Dim rows are undone.
4. Push **DATA** (or press **S2**) to go to that step.

Controls

Control	Action
SHIFT + MINUS	Undo
SHIFT + PLUS	Redo
DATA turn (History page)	Scroll the list
DATA push or S2 (History page)	Go to the selected step
S1 (History page)	Undo one step
S3 (History page)	Redo one step

Tips and limits

- **Track create cannot be undone** by the stock history. KAOSAMPLE refuses a redo that would cross it and shows “Redo not possible”.
- **Delete track cannot be undone.** KAOSAMPLE refuses (“Cannot undo - Delete Track: use Snapshots”).
- **Explode** is only partly undoable (the copied notes go back, the empty tracks stay).
- **EQ, Loudness, Color and Duck settings are not in this list.** Color has its own undo (chapter 4.3).
- **Sample recording can be undone** on the Sample Record screen (SHIFT + MINUS / SHIFT + PLUS). Undo waits about 1.2 s after you stop, and refuses while a recording is still armed or running (“tap the pad to stop first”).
- The stock history holds about 512 steps. Old steps scroll off.
- Insert effects: Undo does not reliably remove the effect. Use ERASE on the insert row (chapter 4.10).
- If the stock app shows a red “Command failed” box, press SCREEN 1, 2 or 3 to clear it. This can happen if you redo past a track create with SHIFT + PLUS.

4.5 Snapshots and recovery

What it does

Every 5 minutes KAOSAMPLE saves a **copy of your project to the card** in `KAOSAMPLE/snapshots`. It keeps the newest **30**. If the MPC's own storage goes bad, a project is lost, or you delete a track, you can bring a snapshot back **from the MPC itself**.

Step by step: restore a snapshot

1. Open **KAOS TOOLS > Snapshots**.
2. The list shows snapshots newest first, with date, time, number and size. The top shows how many seconds ago the live project was saved.
3. Turn **DATA** to pick one.
4. Push **DATA** (or press **S3**) once. The page says "Press again: restore #N".
5. Push **DATA** (or **S3**) **again within 5 seconds** to confirm.
6. The screen goes black for about **35 to 40 seconds** while the app restarts. Wait. Your restored project loads.

Before it restores, KAOSAMPLE saves a **safety copy** of your current project in the same folder (the name ends in "-pre"), so you can go back.

Controls

Control	Action
DATA turn	Scroll snapshots
DATA push or S3, twice within 5 s	Restore the selected snapshot
S1	Refresh the list

Tips and limits

- The app must have been **running for at least about 2 minutes** (125 seconds). If not, the page says "wait N s". Two restarts less than 2 minutes apart make KAOSAMPLE fall back to the stock app.
- If a restored project's samples are missing, the stock app shows "Sample not found" with ABORT and RESOLVE. This is the stock message.
- A restore replaces the open project. Copies of your work are in the "pre" file.
- Snapshots are plain project files. You can also copy one off the card to your computer.

4.6 Step edit

What it does

Edit notes on a grid, with copy, cut, paste, duplicate, a bar menu, zoom and a "stamp" for fast hi-hat rolls.

- **Track View**: tracks are the rows.
- **Pad View**: the pads of the current bank are the rows. Banks follow the stock PAD BANK, and SHIFT + DATA turn or K1 walks through all 16 pads of the bank. The sample name shows inside each row.

Open them from the first menu (**SHIFT + SCREEN 3**): **Track View** or **Pad View**.

Step by step: edit notes

1. Open **Track View** or **Pad View**.
2. Turn **DATA** to move the cursor along the grid.
3. Hit a **pad** to add or remove a note at the cursor (toggle). Push **DATA** to select the note under the cursor.
4. With notes selected, use **K1** to nudge the time, **K2** for length and **K3** for velocity.
5. Press **16 LEVELS** to quantize the selection (to 1/16).
6. **ERASE** clears the selection.

Step by step: copy, cut, paste

1. Select notes.
2. **CHOP** copies. **SHIFT + CHOP** cuts.
3. Move the cursor. **LOOP** pastes. **SHIFT + LOOP** duplicates right after the selection.

Step by step: bar menu

1. Move the cursor into the bar you want.
2. Press **SAMPLE SELECT**. The menu lists: Copy bar, Paste bar (replace), Insert empty bar, Insert copied bar, Delete bar.
3. Turn **DATA** to pick. Push **DATA** to run it.
4. **ERASE** or **SAMPLE SELECT** closes the menu.

Step by step: stamp a roll

1. Hold **NOTE REPEAT**.
2. Turn **K2** to set the rate (1/4, 1/8, 1/16, 1/32, 1/64). The page shows “Rpt 1/n”.
3. Hit a pad. Notes of that pad fill from the cursor to the end of the bar at that rate.

Controls

Control	Action
DATA turn	Move the cursor
DATA push	Select the note under the cursor
SHIFT + DATA push	Select the whole row
SHIFT + DATA turn	Move the row (Pad View: pad row; Track View: track). The selection stays and can span tracks
Pad hit	Toggle a note at the cursor
K1 / K2 / K3	Nudge / length / velocity of the selection

Control	Action
SHIFT + K1 / SHIFT + K2	Nudge / length in steps of 1/16
16 LEVELS	Quantize the selection
ERASE	Clear the selection
CHOP	Copy
SHIFT + CHOP	Cut
LOOP	Paste at the cursor
SHIFT + LOOP	Duplicate after the selection
SAMPLE SELECT	Open or close the bar menu
PLUS / MINUS	Zoom in / out: 1/64, 1/32, 1/16, 1/8, 1/4, 1/2 and 1 bar. At the widest it shows 16 bars
NOTE REPEAT held + pad	Stamp notes
NOTE REPEAT held + K2	Stamp rate
SHIFT + ERASE	Explode the current track (chapter 4.8)

Tips and limits

- **4/4 only.** The bar tools assume four beats to the bar. Quantize still snaps to 1/16, not the zoom column.
- Selection and clipboard hold up to **256 notes**. Bar tools refuse tracks with more than **4000 events**. The grid shows the first 512 events of a track.
- Paste skips a note that already exists at the same time and pitch, and drops notes past the sequence end. The popup says “some skipped”.
- Most jobs are one undo step. A cut or other multi-note edit that was done before the grouping may take several presses.
- Pad hits recorded live show up on the grid after a short delay. A “ghost” note shows right at the hit and is replaced when the real note arrives.
- **Overwrite recording** now clears all old notes the loop passes over (it used to clear only one). Press REC a second time to punch out.

4.7 Track tools: solo, names, move, delete, mute scenes

What it does

- **Solo** a track.
- **Rename** a track by typing on the pads.
- **Move** a track up or down the list.
- **Delete** a track.
- **Mute scenes**: save and recall up to 8 mute patterns per sequence.
- **Recorded mutes**: record mute changes into the sequence while you record, and play them back.
- **Paged track grids**: 16 tracks per page, so you can reach all 32 tracks.

Step by step: Tracks Pro

1. **SHIFT + SCREEN 3 > KAOS TOOLS > Tracks Pro**. You see all tracks, with flags M (mute), S (solo), R (record).
2. Turn **DATA** to pick a track.
3. **S1** mutes, **S2** solos, **S3** renames.
4. **SHIFT + DATA turn** moves the track up or down.
5. **ERASE twice within 4 seconds** deletes the track. The first press shows "ERASE again: delete NAME".
6. Push **DATA** for a track menu: Rename, Move up, Move down, Move to top, Move to bottom, Delete, Clear all solo, Mute scenes, Close.

Step by step: rename (name editor)

1. Press **S3** on Tracks Pro (or pick Rename in the menu).
2. Type on the pads. **PAD BANK A to D** gives four pages of 16 characters:
   - A: letters A to P
   - B: letters Q to Z and digits 0 to 5
   - C: 6 7 8 9 and the symbols `space - _ . + # & ' ! ( ) /`
   - D: the symbols `: , @ $ % * = ? < > [ ] ~ ^ | "`
3. **SHIFT + pad**, or **PAD BANK E to H**, types lower case.
4. **DATA turn** moves the cursor. **ERASE** is backspace. **SHIFT + ERASE** clears the name. **S2** types a space. **PLUS / MINUS** step the character under the cursor.
5. **S3** or **DATA push** accepts. **S1** cancels.
6. Names up to 20 characters. Duplicate names get a number ("Kick 2").

Step by step: mute scenes

1. **KAOS TOOLS > Mute Scenes**. The page shows the 4x4 pads.
2. The **bottom two rows** are **scenes 1 to 8** for the current sequence. Hit a pad to **recall** a scene. **SHIFT + pad stores** the current mutes into that scene.
3. The **top two rows** are mute pads 9 to 16 for the tracks of the current page. Hit a pad to mute. **SHIFT + pad** solos.
4. **S1** arms **REC MUTES**. While the sequencer records, every mute change is recorded.
5. **S3** turns recorded-mute playback on or off for this sequence.
6. **S2** unmutes all. **ERASE twice** clears this sequence's recorded mutes. **SHIFT + ERASE** clears the last scene.

Controls: Tracks Pro

Control	Action
DATA turn	Select a track
SHIFT + DATA turn	Move the track
S1 / S2 / S3	Mute / Solo / Rename
Pad	Select a track (16 per page: pads = tracks 1 to 16, then 17 to 32)
SHIFT + pad	Mute that track
ERASE twice (4 s)	Delete track
DATA push	Track menu

Control	Action
SHIFT + ERASE	Explode the track

Controls: Mute Scenes

Control	Action
Pad, bottom two rows	Recall scene 1 to 8
SHIFT + pad, bottom two rows	Store current mutes as that scene
Pad, top two rows	Mute track 9 to 16 of the page
SHIFT + pad, top two rows	Solo that track
S1	REC MUTES on or off
S2	Unmute all
S3	Recorded-mute playback on or off
DATA turn	Select a track
DATA push	Mute the selected track
ERASE twice	Clear recorded mutes for this sequence
SHIFT + ERASE	Clear the last scene

Controls: Tracks and Track Mute grids

Control	Action
PAD BANK, hold	Tracks grid. A pad selects that track
SHIFT + PAD BANK, hold	Track Mute grid. A pad toggles mute for that track
SHIFT + PAD BANK, tap	Create a new track
K1	Page (16 tracks per page). The page shows “Page n/m”

Tips and limits

- **Delete cannot be undone.** Restore with Snapshots. Delete refuses the last track, and refuses a track that **shares its kit** with another track. The message says “kit shared with N track(s): save+reload”. Save and reload the project, then delete.
- **Solo works per program (kit).** Tracks that share a kit solo together.
- Renaming a track that shares a kit with another track changes the name without an undo step.
- **Mute scenes, recorded mutes, slices and chord sets are saved with the project** (in `kaos-sidecar.txt` next to the project) and copied to the card. They come back when the project loads.
- Pad mutes are recorded for the **current track only**. Scenes are track mutes only.
- **32 tracks maximum.** With full kits, 32 tracks used about 1.7 GB of memory in the lab. The real RAM of the unit is recorded in `mem-probe.txt` (chapter 5).
- Track create cannot be undone (the stock history refuses it).

4.8 Explode

What it does

Splits the current drum track into several tracks so you can mix or mute the parts separately. The new tracks are named after their samples. The source track is kept and **muted** (default). (check on device) the exact split rule (the notes describe the result as "N tracks" created from the track).

Step by step

1. Select the track you want to explode.
2. On the Mixer, Track View, Pad View, Seq Ops or Tracks Pro page, press **SHIFT + ERASE**.
3. A toast says "Exploded N tracks / source muted". If there is not enough room (32 tracks max), it says so instead.

Controls

Control	Action
SHIFT + ERASE	Explode the current track (Mixer, Track View, Pad View, Seq Ops, Tracks Pro)

Tips and limits

- **Only partly undoable.** Undo puts the notes back, but the empty new tracks stay. Delete them in Tracks Pro or restore a snapshot.
- Counts toward the 32-track maximum.

4.9 Sample tools: chop, slices, tail, reverse, extract

What it does

Work on the sample of a pad **without ever changing the WAV file**.

- **Chop** a sample into up to **128 slices**, with a waveform view. Chop by equal regions or by detecting hits. Markers can snap to zero crossings.
- **Extract** a slice to a pad. The pad plays that slice of the same sample (start and end only, no new file).
- **Reverse** a pad.
- **Tail:** the stock **Tail Length**, which is a **looped tail**, not a fade.
- **Fade:** makes a **faded copy** as a new file with a chosen fade out and fade in.
- **Copy pad:** copy a pad (sample, start, end, reverse, tail, levels) to another track.
- The **sample name shows under each pad** on the Pad View, Pad Mixer, Pad Sound and Pad tools pages.

Open **KAOS TOOLS > Sample Tools**. It shows the **Slice view** of the last pad you hit. The same page has **Pad tools**.

Step by step: chop a loop

1. Hit the pad whose loop you want to chop.
2. Open **Sample Tools**.
3. Push **DATA** and pick **Chop Regions N** (equal pieces) or **Chop Threshold** (finds hits).
4. For regions, set the count with **K1** (2, 4, 8, 16, 32, 64, 128). For threshold, set sensitivity with **K2** and the smallest allowed slice with **K3**.
5. Zoom with **PLUS / MINUS** to adjust markers.
6. **S1** adds a marker at the cursor. **S2** toggles move mode: DATA then moves the nearest marker. **S3** deletes the nearest marker (it merges with its neighbor).

Step by step: extract slices to pads

- **One slice to one pad:** hit a pad to select slice number (bank x 16 + pad). Then **SHIFT + pad** sends the selected slice to the pad you hit.
- **All slices:** turn **SHIFT + DATA** to choose the destination start pad. Push **DATA** and pick **Extract ALL to pads**. Pick **Extract ALL (overwrite)** to replace pads that already hold samples.
- **One slice to a chosen pad:** use **Extract slice to Dest** in the menu.

Step by step: Pad tools (reverse, fade, copy)

1. In the DATA menu pick **Pad tools**, or push DATA from Pad tools to go back to the Slice view.
2. Turn **DATA** or hit a pad to select a pad.
3. **S1** flips Reverse. **S2** makes a fade copy (set the fade out with **K2** from 5 to 1000 ms, and the fade in with **K3** from 0 to 50 ms). **S3** starts Copy: choose the destination track with **DATA**, then hit the destination pad. **K1** sets the stock Tail Length.

Controls: Slice view

Control	Action
S1 / S2 / S3	Add marker / Move nearest marker / Delete nearest marker
DATA turn	Move the cursor
SHIFT + DATA turn	Set the destination start pad for Extract ALL
DATA push	Menu: Chop Regions N, Chop Threshold, Link slices ON/OFF, Snap to zero ON/OFF, Extract ALL to pads, Extract ALL (overwrite), Extract slice to Dest, Clear slices, Pad tools, Close
K1	Number of regions (2, 4, 8, 16, 32, 64, 128)
K2	Threshold sensitivity, 0 to 100%

Control	Action
K3	Chop guard: minimum slice length, 2 to 250 ms
PLUS / MINUS	Zoom in and out (up to 16384 times; never below 256 frames)
Pad	Select slice number (bank x 16 + pad). The cursor jumps to its start
SHIFT + pad	Extract the selected slice to that pad (undoable; overwrites that pad)
ERASE twice	Clear all slices (back to one slice, the whole file)

Controls: Pad tools

Control	Action
DATA turn or pad	Select a pad
S1	Reverse (all layers with a sample). One undo step
S2	Fade: render a faded copy as `name_fadeN.wav` and load it on the same pad
S3	Copy pad: DATA picks the destination track, a pad hit picks the destination pad
K1	Stock Tail Length (looped tail)
K2 / K3	Fade out / fade in time (used by S2)
DATA push	Open the Slice view for this pad

Tips and limits

- **Link slices ON:** slices touch each other, and moving a marker moves the end of the slice before. **OFF:** each slice owns its start and end, and Threshold chop also ends each slice where the hit falls silent.
- **Snap to zero** moves each marker to the nearest zero crossing within 2 ms to avoid clicks.
- Chop guard: at most 128 slices, never shorter than the minimum, and a chop that would break the rule is refused (the old slices stay).
- **The stock Tail is a loop tail, not a fade.** Use Fade for a fade. Fade **adds a sample to the project** (a new file named `name_fadeN.wav`).
- **Extract writes layer 1 only.** A pad with samples on layers 2 to 4 keeps them.
- **Slice sets are saved with the project** (sidecar file) and also kept on the card (24 sets). Loading the project brings them back.
- The Slice view reads the WAV from disk. A missing or odd-format file shows “wav not readable”.
- Detection is tuned on test hits. Noisy loops need Sensitivity and Min length tweaks.
- Slice sets are matched by file name and length. Two different WAVs with the same name and length would share one set.

- Undo for Fade is the stock Load Sample step. It undoes the pad assignment but leaves the new sample in the project.

4.10 Pad sound editing: envelopes, filter, choke, layers and inserts

What it does

Pad Sound is one page for one pad. It covers level, pan and tune; the amp envelope; the filter and its envelope; **choke groups**; up to **4 velocity layers**; and a menu of resets and tools. The separate **FX** page adds up to **4 insert effects per pad** from the stock catalogue of 97 effects.

Every edit is a normal stock undoable change. Drum tracks only.

Step by step: shape one pad

1. Hit the pad you want to edit.
2. **SHIFT + SCREEN 3 > KAOS TOOLS > Pad Sound.**
3. Turn **DATA** to pick a row. Turn **K1, K2, K3** to set the row's values (the names are at the bottom).
4. **SHIFT + DATA turn** goes to the previous or next pad. A pad hit also selects that pad.
5. The header shows the pad, kit and sample name. Values still waiting are orange.
6. **S2** resets the current row to the stock default.

The rows

Row	K1	K2	K3
Pad	Level	Pan	Tune (plus or minus 12 semitones)
Amp AHD	Attack	Hold	Decay
Amp SR	Sustain	Release	Decay mode
Filter	Type	Cutoff	Resonance
FltEnv	Amount	Attack	Decay
FltEnv2	Hold	Sustain	Release
FltMode	(check on device)		
EnvType	Amp envelope type	Filter envelope type	-
Choke	Target 1	Target 2	Target 3
Choke 2	Target 4	Mutual	Learn
Layer n "vel"	Vel from	Vel to	Semitone

Row	K1	K2	K3
Layer n “mix”	Fine tune	Level	Pan

EnvType shows **AD** or **AHDSR** for each of the two envelopes. The knob order on the FltMode row is not listed in the notes.

Filter types (shown by name): Off, *Low 1, 2, 4, 6, 8; High 1, 2, 4, 6, 8; Band 2, 4, 6, 8; BS (bandstop) 2P, 4P, 6P, 8P; BB (bandboost) 2P, 4P, 6P, 8P; Model 1, 2, 3; Vocal 1, 2, 3; MPC3000. The “Off*” name is inferred.

Step by step: choke groups

1. Pick the **Choke** row. Use the knobs to set up to 4 pads that this pad will **mute** when it plays.
2. Or push **DATA** and pick **Choke learn**, then **hit the pads** you want as targets. Hit a pad again to remove it.
3. Pick **Choke mutual** to make both pads mute each other. **Clear choke** removes the choke.
4. **ERASE** cancels learn mode.

Step by step: velocity layers

1. Push **DATA** and pick **Split layers 2 / 3 / 4 way** to divide the velocity range among layers. **Layers full range** puts every layer on 0 to 127.
2. Pick a layer row. Set **Vel from / Vel to**, the **semitone**, **fine**, **level** and **pan**.
3. To fill an empty layer, pick **Layer sample from pad** (the source is another pad’s sample) or **Layer sample from pool** (browse the project’s samples, type a name filter on the pads, push DATA to assign).

Step by step: insert effects

1. **KAOS TOOLS > FX.**
2. The page shows the selected pad (a pad hit selects it) and 4 insert rows.
3. Push **DATA** on a row to open the list of effects.
4. **K1** picks the family: All, Delay+Reverb, Dynamics, Filter+EQ, Drive+Misc, Modulation, Vintage.
5. Type on the pads to filter by name. **ERASE** is backspace. **SHIFT + ERASE** clears the filter.
6. Turn **DATA** to scroll. Push **DATA** to assign.
7. **ERASE** on a row clears that insert. **S1, S2, S3** bypass an insert (check on device which button belongs to which slot).
8. The FX page also has links to the stock **Color Compressor**, the **Loudness** page and the **EQ**.

Controls: Pad Sound

Control	Action
DATA turn	Select a row

Control	Action
SHIFT + DATA turn	Previous or next pad (128 pads)
K1 / K2 / K3	The row's three values. A turn applies live while you move it. The change closes into one undo step 220 ms after the knob rests
S1	Mute the pad
S2	Reset the current row to the stock defaults
S3	Solo the pad
Pad hit	Select that pad (it still plays). In choke learn: toggle a mute target. In layer assign: its sample goes to the layer
DATA push	Menu: Reset amp envelope, Reset filter + env, Reset all sound, Split layers 2 / 3 / 4 way, Layers full range, Layer sample from pad, Layer sample from pool, Choke learn, Choke mutual, Clear choke, Close
ERASE	Close the menu; cancel assign or learn

Controls: FX

Control	Action
Pad hit	Select the pad
DATA push	Open the effect list for the selected row
K1 (in list)	Effect family
Pads (in list)	Type the name filter
DATA turn (in list)	Scroll
DATA push (in list)	Assign the effect
ERASE (on a row)	Clear that insert
S1, S2, S3	Bypass (check on device which slot each button controls)

Tips and limits

- **Hold, Sustain and Release** need the **EnvType** row set to **AHDSR**. In the default "AD" type they do nothing. **Release may still not be audible** on drum pads (check on device). Hold and Sustain were proven in the emulator.
- On hardware, **Hold, Sustain and Release were silent on drum pads in v0.20**; v0.21 added the EnvType fix and it is emulator-tested only.
- Choke has 4 slots per pad. Learn refuses a fifth. Choke between pads of different kits is not possible.
- Layers: only layers that hold a sample are editable, plus the next empty one.
- **Only pad inserts work.** Track and master inserts do nothing on this unit.
- Insert **parameters are not editable**. Inserts run with their default settings. The page cannot read back what an insert slot holds from an earlier session; it shows "(active, set before)".
- **Undo of an insert is unreliable.** Use **ERASE** to remove it.

- Inserts are saved in the project and came back after a power cycle in the emulator.
- About 0.3% of the audio time budget is used by the chain in the emulator.

4.11 Stems

What it does

Exports **every track and the master as separate WAV files** to the card, all the same length, so they drop into a DAW lined up. Files are 24-bit, 44.1 kHz stereo.

There is no per-track export in the stock MPC Sample, so KAOSAMPLE makes stems by muting the other tracks and rendering one track at a time. It has two ways to render:

- **Offline:** fast, through the stock Song Export. About 1 second per 21 seconds of audio. It respects your mutes but **does not include** KAOSAMPLE's EQ or Loudness.
- **Real time:** plays the song once per file and records the output. Slower (about the song length per file) but can include the EQ and Loudness on the master.

Step by step

1. Make sure the sequence you want is the current one, and that the card has free space. The page refuses if there is not enough room.
2. **Bypass the stock Color Compressor** first: open **KAOS TOOLS > FX**, push **DATA** on the Color Compressor row, and press **S3** (Bypass). Otherwise stems will not add up to the master. Press the **SAMPLE** button to leave. (check on device) that the compressor shows as bypassed.
3. Open **KAOS TOOLS 2 > Stems**.
4. Set the rows (see below).
5. Press **S1** to start.
6. The page shows the pass number, the file, a progress bar and the time left. The panel is locked while it works.
7. To stop early, press **SCREEN 1** or **PAD FX**. Files already finished stay. The file being made is deleted.
8. When it ends your mutes are restored.

Files appear in **`KAOSAMPLE/stems/<project>-<date>-<time>/`** as `T01-<track name>.wav`, `T02-...`, and `master.wav`, with a `render.txt` that lists frames, range, engine and the peak of each file.

Controls

DATA turn picks the row. Use K1, K2, K3 to change the value. (check on device) exactly which knob changes each row; the notes name the rows and say K2 and K3 set the bar range.

Row	Choices
Content	Stems + master, master only, stems only
Range	Whole sequence, or bars a to b (K2 and K3)

Row	Choices
Master	Dry, or after EQ + Loudness
Tail	0 to 10 seconds
Engine	Auto (offline), or real time

Button	Action
S1	Start
SCREEN 1 or PAD FX	Cancel while rendering

Tips and limits

- **Do not mix offline and real-time files.** They differ by about 4 dB.
- The stock export adds a fixed **3 second tail** and always starts at bar 1. Longer tails come from adding empty bars. A bar range is cut from the full render.
- Choosing "after EQ + Loudness" for the master forces the real-time engine.
- Stems include **Color** but not the duck or the reference.
- **If you already have a stock song export named "Song"** sitting in the Export folder, the page refuses to start. Move or rename it.
- If the app is killed in the middle, the track mutes can stay changed. Restore them in Tracks Pro or from a snapshot.
- Tracks that share one kit retrigger each other's pads (stock behavior), so the master can have fewer voices than the sum of stems.
- Real-time renders can differ by a frame here and there between passes.

4.12 Song to Seq

What it does

Takes a **song** (your list of sequences played in order) and **bounces it into one new sequence**. Every note, with all repeats, lands in one long sequence. It is one undo step, and the result is checked by a count and a checksum of all notes.

Step by step

1. Build your song in the stock Song page.
2. Open **KAOS TOOLS 2 > Song to Seq**.
3. Press **S1**.
4. The page tells you the result and checks. If any note was lost, it says so.

Controls

Control	Action
S1	Bounce the song to a new sequence

Tips and limits

- Refuses if the song is not 4/4, if a track has over 4000 events, if the total is over 40,000 events or 999 bars, or if there is **no empty sequence slot**. The popup says which.
- **Undo is one step.** The new slot goes back to a plain copy of the first song step. The slot stays marked as used, so delete or reuse it yourself.

4.13 Auto-kit

What it does

Finds the hits in a **loop** (or in a **folder of WAVs**), sorts them into **kick, snare, hat and other**, and builds a **16-pad kit**. A preview lets you correct anything before it commits.

Step by step

1. Open **KAOS TOOLS 2 > Auto-kit**.
2. **Source:** *Loop* uses the sample of the **last pad you hit**, so hit the loop pad first. *Folder* uses the WAVs in `KAOSAMPLE/kits/<name>`, `KAOSAMPLE/sounds` or `KAOSAMPLE/sounds/<name>` (up to 128 files).
3. Set **Sens** (how many hits it finds) and **Min hit** (the shortest hit).
4. Pick a **Kit** layout and the **start bank**.
5. Press **S1** to analyse.
6. On the preview, check each hit. Turn **DATA** to pick one. Change its class with **K1** (kick, snare, hat, perc, other) or push **DATA** for the next class. **S2** skips or unskips a hit.
7. Press **S1** to **commit** and build the kit. Hit the pads to audition (there is no audition on the page).
8. **S3** goes back to setup. **Undo** reverses the whole kit in one press.

While it analyses a folder, **SCREEN 1** cancels.

Controls: setup page

DATA turn picks the row; K1 and K2 change the value.

Row	K1	K2
Source	Loop or folder (check on device)	
Folder	Pick the folder (check on device)	
Sens / Min hit	Sensitivity 0 to 100%	Minimum hit length 10 to 250 ms
Kit	Layout	Start bank, A to H (default A)

Button	Action
S1	Analyse

Kit layouts (pads for kick / snare / hat / other): 4/4/4/4, 3/3/6/4, 5/5/3/3, 2/2/4/8, 6/4/4/2, 8/4/2/2.

Controls: preview page

Control	Action
DATA turn	Select a hit
K1	Override the class of the hit
DATA push	Next class for the hit
K2 / K3	Layout / start bank (re-places at once)
S1	Commit: build the kit
S2	Skip or unskip the hit
S3	Back to setup

Tips and limits

- Default placement: kicks on 1 to 4, snares (and claps) on 5 to 8, hats (and cymbals) on 9 to 12, perc and other on 13 to 16 of the start bank. Inside a group, pads are ordered by brightness (hats by length, closed to open).
- The class guess is about **88% right on one-shots** and **82% on loop hits**. Toms, rims, shakers and vocals can land anywhere. **Always check the preview.**
- Overlapping hits, clap double-bursts and open-hat decays can cause misses and extras. Sens and Min hit are your knobs.
- Two hits at the same instant (kick and hat on one 16th) count as one.
- A hit ends at the next hit or its audible tail, so an open hat closed by the next hit is cut there.
- Commit **overwrites the pads it places** on the current track (Undo restores them). Pads that get no hit keep what they had.
- Folder mode analyzes only the first 0.8 seconds of each file, is not recursive, and takes 16-bit, 24-bit or float WAV. The card files are not changed; the stock load copies them into the project.
- In loop mode the commit is refused if the source pad changed since you analysed.

4.14 Groove

What it does

Changes the **feel** of the current track.

- **Swing:** 50% (straight) to 75%, on a 1/8 or 1/16 grid.
- **Humanize:** random timing and velocity changes. The same seed gives the same result.
- **Groove extract:** reads the feel out of a 1 or 2 bar **loop sample**.

- **Groove apply:** puts that feel on the track, with a strength knob.
- **Groove library:** save, load and delete grooves as files on the card.
- **Live:** lets you change values freely without stacking changes on top of each other.

Everything acts on the **current track**. One Undo press reverts a whole apply.

Step by step: swing

1. Open **KAOS TOOLS 2 > Groove**. Pick the **Swing** row.
2. **K1** sets the amount, **K2** the grid (1/8 or 1/16).
3. **S1** applies it to the track.

Step by step: copy the feel of a loop

1. Hit the pad that has your loop. The loop must be **exactly 1 or 2 bars** and start on the downbeat.
2. Pick the **Extract** row. **K1** sets 1 or 2 bars. **K2** sets sensitivity.
3. Press **S1** to extract. The page draws the feel: green bars for velocity, yellow for late, blue for early.
4. Pick the **Groove** row. **K1** sets strength. **S1** applies.
5. **S2** saves the groove to the card. **K2** picks a saved groove. **S3** loads it.

Controls

DATA turn picks the row. SHIFT + DATA turn picks the previous or next track.

Row	K1	K2	K3	S1	S2	S3
Swing	Amount 50 to 75%	Grid 1/8 or 1/16		Apply		
Humanize	Timing 0 to 40 ms	Velocity 0 to 40	Seed 0 to 127	Apply		
Extract	Loop length 1 or 2 bars	Sensitivity		Extract from the last-hit pad		
Groove	Strength 0 to 100%	Pick a saved groove		Apply	Save	Load
Live	On or off			Back to the original		

On the Groove row, **ERASE twice deletes** the picked groove file.

Tips and limits

- The grid for extract and apply is **fixed at 1/16**.
- **Applying swing twice stacks** (as any swing does). Use **Live**, or undo.

- **Live mode** keeps a copy of the original notes. While it is on, every change re-applies from the original. Each settled change is one undo step. It switches itself off if the track or sequence changes, or the notes were edited elsewhere. S1 on the Live row restores the original exactly.
- Grooves live in `KAOSAMPLE/grooves/` as small text files. A name that exists gets "-2", "-3". Nothing is overwritten.
- Acts on **all notes of the track** (no selection or bar range). Notes only.
- Up to **4000 events** per track. Assumes 4/4.
- Tested with test loops, not with real drum loops. Treat results with care.
- Live mode is not true real-time swing. It re-applies from the stored original.

4.15 Retro Rec (retrospective record, tap tempo, Save version)

What it does

- **Retro record:** KAOSAMPLE secretly remembers your pad hits. Press **SHIFT + SAMPLE RECORD** and the last 1, 2, 4 or 8 bars you played drop into the current track, quantized, **even if you were not recording**.
- **Tap tempo: SHIFT + TAP TEMPO** taps a tempo. Plain TAP TEMPO is still the stock button.
- **Loop BPM:** estimates the tempo of the loop on the last pad you hit and sets it.
- **Save version:** one press saves `name-v1`, `name-v2` and so on. Nothing is overwritten.

Step by step: retro record

1. Play pads. The sequence can be playing, or stopped.
2. Press **SHIFT + SAMPLE RECORD** on any screen. The notes land in the current track.
3. For more control open **KAOS TOOLS 2 > Retro Rec**, pick the **Retro** row, then set bars, quantize and source. Press **S1** to drop. **S2** clears the memory.
4. **Undo** removes the whole drop in one press.

Step by step: tap tempo

1. Tap **SHIFT + TAP TEMPO** at least 3 times in rhythm. A popup shows the BPM on each tap.
2. About 1 second after the last tap, the sequence tempo is set once.
3. On the page, **S1** (or DATA push) taps, **S2** resets, **S3** applies now.

Step by step: loop BPM

1. Hit the pad with the loop.
2. Pick the **Loop BPM** row.
3. **K1** chooses /2, x1 or x2. **K2** chooses a range (75 to 170, 60 to 120, or 100 to 200).
4. **S1** finds the tempo. **S2** sets it.

Step by step: Save version

1. Pick the **Version** row.
2. Press **S1**. KAOSAMPLE drives the stock Save screen for you and types the name. This takes about **15 to 20 seconds**, and the panel is locked meanwhile. Do not touch anything.

Controls

DATA turn picks the row. SHIFT + DATA turn picks the previous or next track.

Row	K1	K2	K3	S1	S2	S3
Retro	Bars: Auto, 1, 2, 4, 8	Quantize: off, 1/8, 1/16, 1/32	Source: this track or all tracks	Drop	Clear the memory	
Tap				Tap (or DATA push)	Reset	Apply now
Loop BPM	/2, x1, x2	Range		Find tempo	Set tempo	
Version				Save version		

Combo (any screen)	Action
SHIFT + SAMPLE RECORD	Retro drop
SHIFT + TAP TEMPO	Tap tempo

Tips and limits

- **Auto** uses the loop length while playing, or the phrase you just played when stopped.
- Only **pad hits** are remembered. Not MIDI keyboard notes, not note repeat, and not 16 Levels notes beyond what the pads map.
- The drop **refuses while the stock recorder is recording** (“stop recording first”).
- **4/4 assumed.**
- Tap tempo works between 30 and 300 BPM, and it is refused while recording.
- Loop BPM is a guess for loops that are not exact bars, and for double or half time. It needs about 4 hits. It likes straight 16th loops; swung or triplet loops are less reliable.
- **Save version** was tuned on the emulator, because it types blind into the stock Save screen. On a real panel it may be faster or slower. Names are **lowercase** with dashes, digits and spaces only (“My_Beat” becomes “my-beat-v1”). If typing goes wrong, a wrongly named project may exist on the card (it never overwrites anything), and the page says “WRONG NAME” or “NOT saved”.
- Whether the stock app uses SHIFT + TAP TEMPO and SHIFT + SAMPLE RECORD for something else is unknown. KAOSAMPLE takes those presses.

4.16 Clean up: purge and mono

What it does

- **Purge unused:** finds samples in the project that no pad uses, lists them with total size, and removes them from the project in one undo step.
- **Make mono:** finds stereo samples that are really mono (or one-sided, or with the channels in opposite polarity) and makes a **mono copy**, saving memory. It points the pads at the copy.

Step by step: purge

1. Open **KAOS TOOLS 2 > Clean Up**. The **Purge** row is on top.
2. Press **S1** to scan. The list shows names, sizes and the total.
3. **K3** scrolls the list.
4. Press **S2 twice within 6 seconds** to purge. The second press re-scans and refuses if the list changed.
5. **Undo** brings the samples back in one step.

Step by step: make mono

1. Pick the **Mono** row.
2. **K1** picks the program (kit). **K2** chooses *near-mono only* (default) or *all stereo*.
3. Press **S1** to analyse. The list shows each sample as MONO, ONE (one-sided) or wide, with its side level.
4. Press **S2** twice to convert.
5. The copies are named `name_mono.wav` and saved in `KAOSAMPLE/work`. The originals are never changed.
6. **Undo** returns the stereo samples and every pad value in one step.

Controls

DATA turn picks the row. SHIFT + DATA turn picks the previous or next track.

Row	K1	K2	K3	S1	S2
Purge			Scroll the list	Scan	Press twice: purge
Mono	Program	Near-mono or all stereo	Scroll the list	Analyse	Press twice: convert

Tips and limits

- Purge refuses when the project has a **program that is not a drum program** (keygroup programs).
- **Purge removes samples from the project only.** Files on the card are not deleted.
- **Make mono** works on **one program** at a time. Other programs keep their stereo originals until you purge.
- The “RAM saved” number is an **estimate**. The real saving needs a hardware check.

- "All stereo" mode folds wide material down to mono, which can lower the level. That is why near-mono only is the default.
- WAV loop and cue chunks are not copied to the mono file.

4.17 Perform: scale, chord, arp and Keys

What it does

Turns the 16 pads of your drum program into a **playable instrument** by copying the source pad's sample onto all 16 pads at different pitches. The source sample is treated as **C**.

- **Scale:** pad N plays scale degree N. 16 scales: Chromatic, Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Harmonic minor, Melodic minor, Major pentatonic, Minor pentatonic, Blues, Whole tone, Diminished (whole-half), Phrygian dominant. Any root and octave.
- **Chord:** each pad plays a chord. Diatonic triads or 7ths, or **your own set** (16 chord shapes, one per pad). Own sets are saved per sequence on the card and in the project sidecar.
- **Arp:** an arpeggiator on a scale kit. Patterns Up, Down, UpDown, Random, AsPlayed. Rates 1/4 to 1/32 plus triplets. Gate, 1 to 3 octaves, latch, chord expand.
- **Keys:** play the pad program from a **MIDI keyboard** over 16, 32 or 48 pads.
- **Arp + Keys:** keys on banks E to G (at the default 48-pad span) and the arp on bank H, both at once.

The pitch changes use the normal pad tuning, so pads play, record and undo like any other. **Everything is one undo step ("Perform setup").**

Step by step: scale kit

1. Hit the pad whose sample you want to use, then open **KAOS TOOLS 2 > Perform**.
2. Pick the **Mode** row. Set **K1** to *Scale*.
3. Press **S2** to take the last-hit pad as the source.
4. Pick the **Key** row. Set the root with **K1**, the scale with **K2**, the octave with **K3**.
5. Go back to **Mode** and press **S1** to **Apply**. (Any value change also re-applies after about 0.6 seconds.)
6. Play the pads. Record them as usual.

Step by step: chords

1. Set **Mode** to *Chord*.
2. **Chord** row: **K1** triads or 7ths, **K2** diatonic or your own set.
3. For your own set use the **Edit** row: **K1** root, **K2** shape, **K3** pad. **S1** saves, **S2** loads, **S3** resets.

Step by step: arp

1. Set **Mode** to *Arp* (the same kit as Scale, with the arpeggiator on).

2. **Arp** row: **K1** pattern, **K2** rate, **K3** gate. **Arp 2** row: **K1** octaves, **K2** latch, **K3** chords.
3. Hold pads in the kit bank. They are arpeggiated on the sequencer grid. The stock recorder records the result.
4. **ERASE** clears held arp pads.

Step by step: Keys

1. Pad MIDI In is turned on for you about 25 seconds after the MPC starts. The **Keys** row shows "PadIn on (boot)" when that worked. If it shows **FAILED**, set **Settings > MIDI > Pad MIDI In = On** by hand. Make sure the right MIDI port is chosen.
2. Set **Mode** to *Keys*, or to *Arp+Keys* to play and arp together.
3. **Keys** row: **K1** base note, **K2** span 16, 32 or 48.
4. Play the keyboard. Notes become pad hits, and the stock app plays and records them.

Controls

DATA turn picks the row.

Row	K1	K2	K3	S1	S2	S3
Mode	Mode: Off, Scale, Chord, Arp, Keys, Arp+Keys	Bank	Octave	Apply	Take the last-hit pad as source	Go to the bank
Key	Root	Scale	Octave			
Chord	Triads or 7ths	Diatonic or user				
Edit	Root	Shape	Pad	Save	Load	Reset
Arp	Pattern	Rate	Gate			
Arp 2	Octaves	Latch	Chords			
Keys	Base note	Span 16 / 32 / 48	Gate cut on / off			

On the Mode row the top bar shows S1 "Apply", and S3 shows "Bank". On the Edit row S3 shows "Reset". ERASE clears held arp pads.

Tips and limits

- A kit **overwrites the pads** it uses (Undo restores). Other layers on those pads are cleared.
- The sample is assumed to be C. Pitches beyond plus or minus 48 semitones clamp.
- **Gate** (Keys row, K3 "cut") makes kit pads stop when their note ends, so it really cuts drum tails. It also applies to the arp gate. It is on by default; set it off to keep one-shot tails. In the lab a 4-second sample stopped after 0.33 s with Gate on.
- **Arp works only while the current bank is the kit bank.**

- **Arp + Keys together:** keys on banks E to G, arp on bank H. The keyboard does not feed the arp, and both use one source sample.
- **Pad MIDI In** is set at boot, so you normally do not set it. A USB keyboard was only tested through the lab's MIDI port, not on a real unit.
- A chord apply writes about 180 values at once. This was fine in the emulator but not stress-tested on the unit.
- In the Arp, the first note waits for the next grid step while playing.

4.18 Looper

What it does

A **multi-pass looper** synced to the sequence. Record a loop, overdub more passes (up to 8), undo a pass, change pass levels, and **export** the result to a pad or into the sequence. It records the **MPC's own output** (a resample of the master), the **line-in**, or both. It hears your pads, sequence and anything else the MPC plays. The loop never records itself, so there is no feedback.

Two looper tiles exist:

- **Loop Rec** on KAOS TOOLS 2 is the new one described here. Its page title is "LOOPER".
- **Looper** on KAOS TOOLS is the older page that drives the stock looper. It is still there and unchanged. (check on device) its controls; the notes for this build do not describe them.

Step by step

1. Press **PLAY** so the sequence runs. **The looper only works while the transport plays**, unless **free-run** is on (see below).
2. Open **KAOS TOOLS 2 > Loop Rec**.
3. Turn **K1** (while the loop is empty) to choose the length: the sequence length, or 1, 2, 4 or 8 bars.
4. Press **S1**. The state shows "WAITING FOR BAR 1", then "RECORDING" when the next loop boundary arrives.
5. After the first pass the looper carries straight on into the next pass (overdub), starting on the same beat. Press **S1** while it is recording to say "stop after this pass". Press it again to keep overdubbing.
6. Press **S2** to undo the last pass. **K1** picks a pass, **K2** sets its level from 0 to 150%. **K3** sets the loop's playback volume.
7. Press **S3** to mute or unmute the loop. Recording keeps going.
8. To keep the loop, turn **DATA** to a row at the bottom and push it:
    - **To pad:** make a 24-bit WAV on the **next empty pad** of the current track.
    - **To Seq (this track):** the same, plus a note at bar 1 on the current track.
    - **To Seq (NEW track):** the same, on a new drum track.
9. Each export is **one undo step**.
10. **ERASE twice** clears the loop.

Source and Run rows

- **Source: K1** picks Master, Line-in or Master+In (both summed). **K2** picks the tap: **Pre** (before EQ, Color and Loudness, as in v0.29) or **Post** (after them). Post adds the loop playback after the chain, so it is not processed twice and not limited.
- **Run: K1** turns **free-run** on or off. With free-run on, the loop keeps looping and recording when the transport stops, and a pass armed while stopped starts at once. **K2** turns **tempo stretch** on or off (on by default).

Each relock to bar 1 is crossfaded, so it does not click.

Controls

Control	Action
K1	Empty loop: length (Seq, 1, 2, 4, 8 bars). Otherwise: select a pass
K2	Level of the selected pass, 0 to 150%
K3	Loop playback level
S1	Record or Overdub. While armed: Disarm. While recording: stop after this pass
S2	Undo last pass. While recording: drop the running pass
S3	Mute or unmute the loop
DATA turn	Pick a row: To pad, To Seq (this track), To Seq (NEW track), Source, Run
DATA push	Run the row
ERASE twice	Clear the loop
SHIFT + DATA turn	Previous or next track (the track that gets the pad or note)

Pads, transport and SHIFT combos pass through, so you can play pads into a pass.

Tips and limits

- **Master source:** records the master resample. With the **Pre** tap, EQ, Color and Loudness are **not** in the loop; with **Post** they are. Master volume is not baked in either way.
- **Needs the transport playing** unless free-run is on. With free-run off, stopping the transport stops the loop and drops a running pass.
- **Tempo change:** with tempo stretch on, the loop is time-stretched to the new tempo once the tempo has been steady for about 0.7 s (pitch is kept). Stretch works from half to double the original tempo. Outside that, or with stretch off, the page warns **“tempo changed: re-record the loop”**. Stretch was tested on tones and noise, not on real drum loops, and artefacts build up over many changes.
- **Meters:** the loop length follows the bar length, so 3/4, 6/8, 5/4, 7/8, 12/8 and 2/4 work. 3/4 and 6/8 count the same way (the page shows 3/4 for both). A sequence that changes meter in the middle is not supported. Only constant meters were tested in the lab.

- A pass starts within about one pulse of the boundary.
- Max **8 passes** (fewer on long loops, to protect memory). Loops up to 16 bars at 60 BPM fit.
- **Export does not stall.** The file is written in the background; the screen only waits for the short stock load step (about 17 ms for a 2-bar loop in the lab).
- Exports are saved in `KAOSAMPLE/work` and in the project's data folder, named like `loop_<bars>b_<bpm>_<time>`. They are never overwritten. Free space is checked on the card and the internal storage.
- Redo of a new-track export can fail when the stock history is full. Nothing is lost. Run the export again.

4.19 Files and templates

What it does

- **Save as template:** keep the whole project, with all its kits, as a template on the card.
- **New from template:** start a new project from a template. The template itself is never changed.
- **Default template:** works from any project. A fresh project (2 tracks or fewer) builds a starter template from your first four kits (`KAOSAMPLE/kits`), at 90 BPM and 4 bars. Any other project is saved as it is, as `kaos-default`.
- **Starter (STRIP):** on a fresh project, removes the stock starter kit and unused samples before the template is saved. In the lab the template went from about 81 MB to about 8.6 MB.
- **New folder:** make a folder on the card or the internal storage, named like the Name row.
- **Random project names:** an adjective plus a noun, such as "dusty-kick" or "warm-loop". It never reuses a name that is taken.
- **Browser jump:** SHIFT + DATA jumps 5, 10 or 20 rows at a time in the stock Sample Select, Project Select and FX Select lists.

Step by step: save a template

1. Open **KAOS TOOLS 2 > Files**. Pick the **Template** row.
2. Pick the **Name** row. **K1** chooses *random* or *from project*. **S1** rolls another name.
3. Go back to **Template**. Press **S1** (**Save**). KAOSAMPLE uses the stock Save screen to type the name and save, then copies the project onto the card. This takes a while and shows a progress line.

Step by step: start from a template

1. **Wait until the MPC has been running for about 2 minutes.** The page refuses earlier ("wait Ns").
2. On the **Template** row, use **K1** to pick the template.
3. Press **S2** twice (**New**). The first press warns you: "replaces the open project (kept as kaos-prev)".
4. The screen goes **black for about 35 to 40 seconds** while the app restarts. Do not turn the MPC off.
5. The new project opens. It is saved as a copy, so the template stays untouched.

Step by step: other rows

- **Folder row: K1** chooses where (KAOSAMPLE, Card Projects, Internal Projects). **S1** creates a folder named like the Name row.
- **Browser row: K1** sets the jump size: off, 5, 10 or 20. Then, in the stock **Sample Select**, **Project Select** and **FX Select** lists, **SHIFT + DATA turn** jumps that many rows.
- **Starter row: K1** keeps the stock starter kit, or STRIPs it (see above).

Controls

DATA turn picks the row.

Row	K1	S1	S2	S3
Template	Pick the template	Save as template	New from template (press twice)	Default template (press twice)
Name	Style: random or from project	Roll another name		
Folder	Where: KAOSAMPLE, Card Projects, Internal Projects	Create folder		
Browser	Jump size: off, 5, 10, 20			
Starter	Keep or STRIP the stock starter kit			

The top bar shows S1 as Save, Roll, Create (depending on the row), S2 as “New” and S3 as “Default” on the Template row.

Tips and limits

- **A template is a full copy**: about **90 MB** with the stock starter kit, or about **9 MB** with STRIP. A new project made from it is about the same size again. The layer refuses when space is short, but it **never deletes your old projects**.
- New from template **restarts the app** and **replaces the open project**. The old one is kept as a `kaos-prev` copy and a safety snapshot. There is no “unsaved?” check. **Save first.**
- **Safety (v0.30.1):** New from template refuses unless the app is healthy. It keeps a safety copy of the new project’s samples on the card (in `KAOSAMPLE/safety`) and repairs the project after the restart if the stock app drops sample files. Old safety copies are pruned (the newest 2 are kept). In the lab, the stock app once deleted sample files from a swapped-in project, for a reason nobody has found yet. **The template folder on the card is never touched, so it is the copy to trust.**
- If the app had already fallen back to stock (two quick exits), the layer is not running after the restart and the job cannot finish.

- Template names must pass the stock editor: lowercase letters, digits, and `- . #`. Typing costs about 0.8 seconds per character, so names stay short.
- Jump works in the stock **Sample Select**, **Project Select** and **FX Select** lists. Other pickers (Sample Save/Export folder chooser, SD Card Access) are not covered.
- **Remember the last folder across a restart is not possible.** The stock app never saves it, and replaying key presses at start-up is not safe.
- **Default template** works from any project. Only a fresh project (2 tracks or fewer) gets the card kits, which must be laid out as `KAOSAMPLE/kits/<kit>/<kit>.xpm`.
- **Kits on the card:** KAOSAMPLE can load a kit from `KAOSAMPLE/kits/<name>/` onto a new track (all 16 pads). (check on device) which page has the load command; the notes only say it is the same kit loader that Default template uses.

4.20 Monitor: CPU and RAM

What it does

A small system monitor on the **About** page. It reads once a second, and only while the page is open.

It shows:

- **CPU** use in percent,
- **RAM** used and total,
- **battery** percent and charging status,
- **temperature**,
- the app's own memory use,
- **Xrun** count: audio dropouts. It turns red when it is above 0.

Step by step

1. **SHIFT + SCREEN 3 > About.**
2. Watch CPU and RAM while you use the heavy tools (Looper, Stems, 32 tracks).

Controls

There are no controls. It is a read-out.

Tips and limits

- **Battery** is unverified on hardware ("no sensor" is shown if the unit has none). Temperature shows "n/a" if there is no sensor.
- **RAM cannot be stretched.** The unit's kernel has no swap or compressed RAM. Track count and loop length are bounded by real RAM.
- **Xrun counter:** only changes during steady playback mean something. The emulator gave start-up dropouts, so the number says little about the hardware yet.
- Every boot writes a `mem-probe.txt` file to the KAOSAMPLE folder with the unit's real RAM size and kernel facts. **Please send this file** so 48 or 64 tracks can be considered (chapter 6).

- A low-memory plan: use Clean Up (4.16) to purge unused samples and make near-mono samples mono.

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5. The card folder map

KAOSAMPLE uses two places on the SD card.

5.1 Top of the card

Item	What it is
`autorun/`	The KAOSAMPLE program itself: the seven files you copied (chapter 2). Delete this folder and the MPC is stock again. Never edit the files.
`KAOSAMPLE/`	Created on first start. Your data and logs. Safe to keep or delete (chapter 2.4).
`MPC-Sample/Projects/`	The stock projects folder on the card. The Files page lists it as "Card Projects" for New folder.

5.2 Inside the KAOSAMPLE folder

Folders

Folder	What is in it	Made by
`snapshots/`	Project copies saved every 5 minutes, newest 30 kept. Named `snap-NNNN-MMDD-HHMM.xpj`. A name ending in `-pre` is a safety copy made just before a restore.	Auto, and Snapshots restore
`stems/`	One folder per render, `<project>-<date>-<time>/`, with `T01-<track>.wav` … `master.wav` and `render.txt`.	Stems
`refs/`	**You** put reference WAV files here for Color's Reference A/B.	You
`kits/`	Card kits, one folder per kit: `kits/<name>/<name>.xpm` plus its WAV files. Also a source folder for Auto-kit and Default template.	You, and the kit saver
`sounds/`	Folders of one-shot WAVs for Auto-kit's folder mode (`sounds/` or `sounds/<name>`).	You

Folder	What is in it	Made by
templates/	One folder per template: <name>.xpj, <name>_[ProjectData]/ (all WAVs) and template.info. A folder without template.info is an unfinished copy and is not listed.	Files
work/	New WAV files made by the tools: fade copies (name_fadeN.wav), mono copies (name_mono.wav), and loop exports (loop_<bars>b_<bpm>_<time>.wav).	Sample tools, Clean Up, Looper
grooves/	Saved grooves, <sample name>.grv, plain text.	Groove
chords/	Your own chord sets, one file per sequence (S<n>.chd). A copy also travels in the project sidecar.	Perform
safety/	Safety copies of a new project's samples made by New from template. The newest 2 are kept.	Files
sidecars/	Card copies of each project's sidecar (mute scenes, slices, chord sets). They come back when the project loads.	Automatic

Files

File	What it is
fx.txt	Saved settings: master EQ, Loudness, and the Color, Duck and Reference-file-name lines. Written about 2 seconds after a change.
fxundo.txt	The Color undo history (32 steps). It comes back after a restart.
usb-probe.txt	A one-time USB probe. It is written once.
slices.txt	Your chop slice sets. 24 are kept. A copy also travels in the project sidecar.
mutes.txt	Mute scenes and recorded mutes. A copy also travels in the project sidecar.
files-pending.txt, files-safety.txt	Short-lived notes used during New from template. They vanish when the job ends.
layer-events.log	When the app started and exited, and the exit codes. **Send in bug reports.**
layer-last-exit.log	What was happening at the last exits and crashes. **Send in bug reports.**
layer-app.log	The recent running log of the app. **Send in bug reports.**
mem-probe.txt	Written at every start: the unit's real RAM size and kernel facts. **Send in bug**

File	What it is
	reports.
`crash-last.txt`	Written after a crash with the details. Send it too if it exists.
`device-content.txt`	A one-time listing of files on the MPC, made for Omar. Not needed for bug reports.

Things that live **in the project** and travel with it: pad sounds, step edits, inserts, tempo, extracted slices (as start and end settings), and the sidecar file `kaos-sidecar.txt` (mute scenes, recorded mutes, slice sets, chord sets). Things that live **only on the card**: grooves, settings in `fx.txt`, the Color undo in `fxundo.txt`. The copies in `sidecars/` are the card copies of the sidecars.

6. Troubleshooting and how to report a bug

6.1 Troubleshooting

Problem	Likely cause and fix
I see only the normal MPC, no KAOSAMPLE splash	The firmware is not 1.3.0.12 (KAOSAMPLE turns itself off). Or it crashed twice within 2 minutes and fell back to stock. Or the `autorun` folder is not at the top of the card, or a file was renamed. Check these, then restart the MPC.
SHIFT + SCREEN 3 does nothing	KAOSAMPLE is not running (see above). Check that the splash showed.
A job says “wait N s” or “refused: app exited N s ago”	Restore, New from template and similar jobs need the app to have been running for about 2 minutes, with no recent exit. Wait, then try again.
The screen is black for about 40 seconds after a restore or New from template	Normal. The app is restarting. Wait. Do not turn the MPC off.
Stock message “Sample not found” with ABORT and RESOLVE after a restore	The restored project is missing samples. This is the stock message.
A red “Command failed” box appears and the buttons stop working	Press SCREEN 1, 2 or 3 to clear it. Avoid redo (SHIFT + PLUS) past a track create.
“Cannot undo - Delete Track” or “Redo not possible”	These are on purpose. Track create and delete cannot be undone. Use Snapshots.
Undo says “tap the pad to stop first” on the Sample Record screen	A recording is still armed or running. Tap a pad to stop it, wait about 1.2 s, then

Problem	Likely cause and fix
	undo. If it was armed and never triggered, leave the screen and come back.
Delete track says “kit shared with N track(s)”	Save and reload the project, then delete again.
Keys mode plays nothing	Look at the Keys row. “PadIn on (boot)” means Pad MIDI In is on. If it says “FAILED”, set Settings > MIDI > Pad MIDI In = On by hand. Check the MIDI port.
Arp does nothing	The current bank must be the kit bank, and the mode must be Arp.
Looper says “WAITING FOR PLAY”	Press PLAY. The looper only runs while the transport plays, unless free-run is on (Run row, K1).
Looper says “tempo changed: re-record the loop”	The new tempo is outside half to double the original, or tempo stretch is off (Run row, K2). Turn stretch on, or record the loop again.
Stems do not add up to the master	Bypass the stock Color Compressor first (4.11). Do not mix offline and real-time files.
Stems refuse to start	Not enough card space, or a stock export named “Song” is sitting in the Export folder. Free space, or move that file.
Auto-kit put a hit in the wrong group	Normal at about 82 to 88% accuracy. Fix it in the preview with K1 before you commit.
Reference A/B says “load a reference first”	Put WAVs in `KAOSAMPLE/refs`, pick one on the Ref row, push DATA to load, then press S1.
Reference sounds quieter or louder than expected	The loudness meter needs a few seconds of music. Look for “peak cap” on the page, and try the Trim knob.
Color knob clicks or zippers on a fast sweep	v0.32 glides Color settings. A very fast sweep is softened on purpose. Tell Omar if it still happens.
Track count stops at 32	That is the maximum.
Out of memory, slowdowns, or dropouts	Use Clean Up: purge unused samples and make near-mono samples mono. Close heavy pages. Check About for CPU, RAM and Xrun.
A save-version name looks odd	The stock editor allows only lowercase letters, digits, dash, dot, hash and space. Names are changed to fit.

Problem	Likely cause and fix
I lost a project	Open Snapshots (4.5). If the unit will not start KAOSAMPLE, copy a file from `KAOSAMPLE/snapshots` on the card to your computer and send it to Omar.
I want to go back to stock	Delete the `autorun` folder from the card (chapter 2.4).

6.2 How to report a bug

After a crash or weird behavior, send Omar:

1. **What you were doing**: which screen and which buttons.
2. **The time it happened**.
3. **These files from the card's `KAOSAMPLE` folder:**
   - `layer-events.log`
   - `layer-last-exit.log`
   - `layer-app.log`
   - `mem-probe.txt`
   - `crash-last.txt` too, if it exists.

To get them: power off the MPC, put the card in your computer, and copy the files out of the `KAOSAMPLE` folder.

Tips for a good report:

- Say which tool (for example "Looper, overdub pass 2").
- Say how long the app had been running when it happened.
- If the screen froze or went black, say for how long.
- If you restored a snapshot, say which one.
- Please do not wipe the logs before you send them. They are small, and they are overwritten by newer events.
- Do not share this build with others.

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7. Known limits

7.1 Hard limits of the MPC Sample

These are walls of the platform, not bugs to be fixed later.

- **Stock inserts on tracks and master do not process audio.** Only pad inserts work. KAOSAMPLE ships its own master EQ, Loudness and Color.
- **No per-track audio.** There is no per-track kill EQ or processing. Stems are made by muting tracks and rendering.
- **No extra memory.** The unit has no swap and no compressed RAM, so memory cannot be stretched. Track count and loop memory are bounded by the real RAM.

tracks maximum. 48 or 64 tracks are not offered until the real RAM size is confirmed from `mem-probe.txt`.
- **No licensing bypass.** KAOSAMPLE does not touch licensing or the boot chain. Features that the stock product locks (for example keygroup programs and MIDI tracks) stay unavailable.
- **The stock app owns undo.** Track create cannot be undone. Delete track cannot be undone. Explode is only partly undoable. EQ, Loudness, Color and Duck settings are not in the stock undo list (Color has its own 32-step undo, kept across restarts). Sample recording can be undone on the Sample Record screen.
- **No remembered browser folder.** The stock app stores no last-folder state, so the file browsers cannot pick up the last folder after a restart.

7.2 Beta status

- Everything from v0.21 up has been tested only in an emulator (see the table in 1.3). Timing, speed and button feel on the real unit are not yet known.
- An earlier display crash seen on hardware was hardened but its root cause was not found. An earlier case where the playhead appeared frozen on hardware was not reproduced; extra logging was added to catch it next time.

7.3 Limits by feature

Feature	Limit
Master EQ	Master only, no per-track EQ
Loudness	Not in undo. Skipped by offline stem renders
Color / Duck / Reference	Loudness meter is LUFS-style, not certified. Duck sources: notes of a set of tracks, live pad hits, MIDI in and arp / Perform hits (each switchable). Very fast sweeps are softened on purpose. Reference list max 48 files
Undo	History holds about 512 steps. Several actions cannot be undone (7.1)
Snapshots	30 kept. Restore needs about 2 minutes of app uptime. About 35 to 40 seconds of black screen
Step edit	4/4 only. Quantize snaps to 1/16. Selection 256 notes. Bar tools refuse tracks over 4000 events. Grid shows 512 events per track
Track tools	Delete not undoable. Delete refuses a shared-kit track. Solo is per kit. Mute scenes and recorded mutes are saved with the project and on the card. Pad mutes record for the current track only
Explode	Only partly undoable. Counts toward 32 tracks

Feature	Limit
Sample tools	Slice sets are saved with the project and on the card (24 kept). Extract writes layer 1 only. Tail is a loop tail, not a fade. 128 slices max
Pad Sound	Drum tracks only. Release may be silent on drum pads. Choke has 4 slots. No effect parameters. Undo of an insert is unreliable
Stems	Fixed 3 second stock tail. Offline skips EQ and Loudness. Offline and real-time differ by about 4 dB. Track mutes can stay changed if the app is killed
Song to Seq	4/4 only. Needs an empty sequence slot. Max 4000 events per track, 40,000 total, 999 bars
Auto-kit	About 88% class accuracy on one-shots, 82% on loops. Folder mode 128 files, not recursive, 0.8 s analyzed per file
Groove	Whole track only. Grid fixed at 1/16. Loop must be exactly 1 or 2 bars. Tested only on test loops
Retro Rec	Pad hits only. Refuses while recording. 4/4 assumed. Save version tuned on the emulator
Clean up	Purge needs drum programs only. RAM saving is an estimate. Make mono works on one program at a time
Perform	Arp only in the kit bank (bank H with Arp+Keys). Keys on banks E to G at span 48. The keyboard does not feed the arp. Pad MIDI In is set at boot; “FAILED” means set it by hand. Gate is on by default. USB keyboard path not yet tested on a real unit
Looper	Master, line-in or both. Pre or post tap. Needs the transport unless free-run is on. Stretch 0.5x to 2x, tested on tones (not drum loops). Meter changes mid-sequence not supported. 8 passes max. Line-in hardware levels and latency untested
Files	Templates are about 90 MB with the stock starter kit, about 9 MB with STRIP. New from template restarts the app and replaces the open project. Jump works in three stock lists. The last folder is not remembered after a restart (the stock app never saves it)
Monitor	Battery reading unverified

7.4 What is not in this build

- The **KAOSAMPLE Companion** Mac app (card status, log reader, kit sync, stems browser, project backups) is a separate project. It is not part of this tester build and has not been used with a real card yet.
- This guide does not cover the Pad Mixer, Sequences, Song, MIDI, Programs and Stock screens tiles, because the notes for this build do not describe their controls.

Thank you for testing KAOSAMPLE. Your reports on the real machine are what turn this beta into a release.